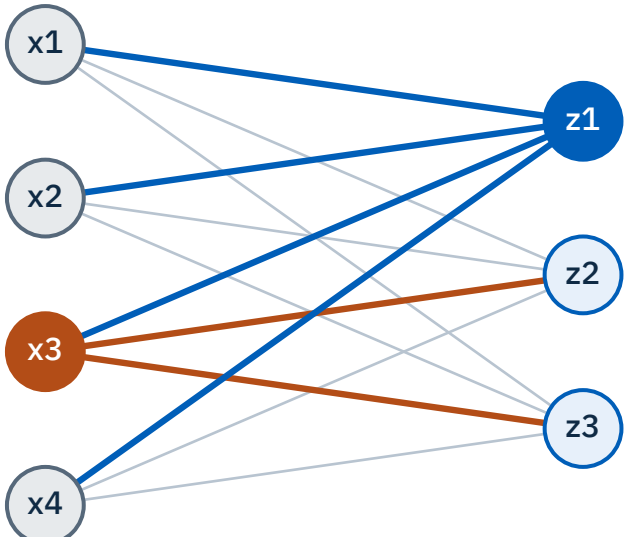

fan_out = 3

units fed by one input

fan_in = 4

inputs into one unit

$$\mathrm{Var}(z) = \mathrm{fan}_{\mathrm{in}} \cdot \mathrm{Var}(w) \cdot \mathrm{E}[x^2]$$

keep Var(z) steady → Var(w) ≈ 1 / fan_in (× 2 for ReLU)